Krish Nagaral

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Skills

Languages: Python, Java, C++, SQL
ML Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Stable Baselines3, SHAP
LLM & Agentic AI: Context Engineering, Prompt Engineering, Function Calling, Tool Use, LangChain, LangGraph, MCP, Hugging Face Transformers, LoRA, PEFT
Retrieval & Vector Search: RAG Pipelines, FAISS, Pinecone, HNSW, BM25
MLOps & Deployment: Docker, Kubernetes, MLflow, Airflow, FastAPI, Flask, gRPC, CI/CD (GitHub Actions), Prometheus, Grafana
Cloud & Data Infrastructure: AWS (EC2, S3, SageMaker, Bedrock, ECS, EKS, Lambda), CloudWatch, PostgreSQL, Redis, Git, Linux
Domain Expertise: Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, NLP, Model Evaluation

Experience

AI/ML Engineer Intern, Coutloot & Flyads.ai, Mumbai — Aug 2024 – Aug 2025
- Improved LLM response accuracy to 86% on evaluation benchmarks by fine-tuning LLaMA and Gemma with LoRA, powering prompt and script generation for the video pipeline
- Scaled the AI backend horizontally to 5,000+ concurrent requests by building Docker-based FastAPI microservices around LLM inference pipelines
- Reached 97% contextual retrieval precision by designing RAG pipelines with vector embeddings and semantic search
- Cut API latency by 38% by adding asynchronous processing, request batching, optimized caching, and better GPU utilization
- Improved system reliability by adding structured logging, centralized error handling, and real-time model monitoring dashboards
- Built the generation backend for an AI video marketing tool: async FastAPI job processing around Sora, with fine-tuned LLaMA models writing the prompts, scripts, and captions for on-demand seller videos
- Optimized Sora video generation with reinforcement learning, using seller engagement as the reward to tune presenter presets
- Connected LLM outputs to backend services through function calling, triggering Sora render jobs and returning finished videos to sellers

Software Tester Intern, Coutloot, Mumbai — Apr 2024 – Jun 2024
- Reached 88% defect detection coverage by designing automated and manual testing workflows for backend services
- Ensured 99.8% production stability at release by running regression testing and performance benchmarking
- Sped up developer fixes by documenting API behavior, edge cases, and failure scenarios with traceable test cases

Software & Cloud Engineer Intern, Tata Communications, Mumbai — Sep 2023 – Mar 2024
- Accelerated release cycles by 60% by implementing AWS-based infrastructure and service orchestration across EC2, S3, and Lambda
- Sustained 99.9% application uptime by supporting EC2, S3, IAM, and VPC based production systems
- Improved IoT-to-cloud pipeline performance by 63% by profiling and tuning bottlenecks across device ingestion and cloud processing
- Supported production incident resolution on live systems by analyzing logs and debugging network failures with cross-functional teams
- Deployed software releases across AWS environments, coordinating with developers to keep test and production setups consistent
- Streamlined onboarding by authoring deployment documentation and training materials

Projects

CodeMentor AI: Multi-Agent Codebase Assistant | GitHub | Live Link | FastAPI, Next.js, RAG, LangGraph, SSE
- Built an agentic codebase assistant by orchestrating 5 LangGraph agents (planner, retrieval, analyst, mentor, memory) across 11 FastAPI endpoints, pairing FAISS vector search with Tree-sitter AST parsing to answer queries under 800ms p95 latency over streaming SSE
- Monitored retrieval precision, latency percentiles, and utilization with Prometheus and Grafana, then ran 3 data-driven tuning cycles that improved answer relevance; shipped with Docker on Render/Vercel and GitHub Actions CI

BoxBox: F1 Pit Stop Strategy Optimizer | GitHub | PPO, Gymnasium, gRPC, Flask, Prometheus, Grafana, Docker, Kubernetes
- Improved mean race finish by 5.7 positions over baseline (P7.8 vs P13.5) by training a PPO agent in a custom Gymnasium environment built from 45,623 real telemetry laps across 42 Grand Prix, with a 1.3-stop strategy matching real F1 patterns
- Served the agent over gRPC at 4.67ms p95 inference (40x under the 200ms target), streamed live recommendations via Flask-SocketIO, and monitored action distribution, latency, and model confidence on a 5-panel Grafana dashboard

Campaign Response Prediction: ML Classification Pipeline | GitHub | LightGBM, XGBoost, PostgreSQL, SHAP, Power BI
- Delivered a 1.9x lift over blanket outreach (top 4 deciles capture 74% of responders at 40% of contact cost) by benchmarking LightGBM, XGBoost, and logistic regression with 5-fold stratified CV and SMOTE, reaching ROC-AUC 0.93 and F1 0.86
- Built a modular CLI with swappable models and data sources, adding SHAP explainability and auto-exported lift charts for Power BI targeting dashboards

Education

Northeastern University, MS Computer Science, AI & ML Specialization — Sep 2025 – May 2027
Vellore Institute of Technology, BTech Computer Science — Aug 2021 – May 2025

Certifications

Anthropic Academy: Claude Code in Action | Claude with the Anthropic API | Claude with Amazon Bedrock | Introduction to Claude Cowork
AWS: Solutions Architect Associate | Machine Learning Associate | AI Practitioner
Other: DeepLearning.AI Machine Learning | DeepLearning.AI Prompt Engineering | Nvidia RAG | Hugging Face MCP | IBM Cybersecurity

Publications

- Comparative Analysis of Machine Learning Algorithms for Heart Attack Prediction, IJFMR, 2024
- Quantum Cryptography: Quantum Key Distribution Protocols for Secure Communication, IJNRD
- Green and Speculative Computing: Energy Efficient Computing Strategies, IJNRD
- Mechanistic Interpretability of Code Correctness in LLMs via SAEs: Reproduction and Extension (under review)
- Future Lens: Extension of Pal et al. (2023) (under review)